Final Project 3: Audio Sampler

Project Description

You will build a simple audio sampler tool which provides basic functionality for audio synthesis and signal processing. In addition to GUI and event-driven programming, this project will expose you to the following concepts:

- Filtering and signal processing in MATLAB.
- Managing audio files in a filesystem.

The following built-in functions will be useful for this project:

- Import an audio file from a folder: `uigetfile()`
- Read an audio file into an array: `audioread()`
- Create an audiofile from an array: `audiowrite()`
- Create an object to play/pause/stop audio: `audioplayer()`

Project Requirements

Your project will be divided into two components: required **core functionality** and **reach features** which expand on the core for a more interesting and creative project. Core functionality is described below. You will have the opportunity to propose your own reach features. An example GUI is provided in Figure 1 below.

Core Functionality (Must-Have)

The following functionality is required:

1. A reasonably attractive and well-organized user interface.
2. A panel for audio file input with a minimum `3×3` grid of buttons where audio files can be loaded and played.
3. An axis for displaying the waveform of the audio that is being played with respect to time.
4. A panel for selecting one of the 9 audio samples and editing that sound file by chopping with a user defined start and end time.
5. A panel for selecting one of the 9 audio samples editing the sound files with effects such as control of the volume, changing playback direction, changing playback speed, or adding delay (echo).
6. A panel for generating a pure tone (single frequency) by using at least two mathematical functions and saving it into one of the audio sample placeholders in the 3×3 grid. Some options are sine wave, square wave, sawtooth and AM modulation wave.
7. An axis for displaying the waveform generated by the synthesizer with respect to time.

Reach Features

The reach features are a way to make your project more interesting and personalized. You should choose and describe at least two reach features in your project proposal. Examples of reach features would include more sophisticated functionality like filtering in the frequency domain and/or visualization of frequency components.

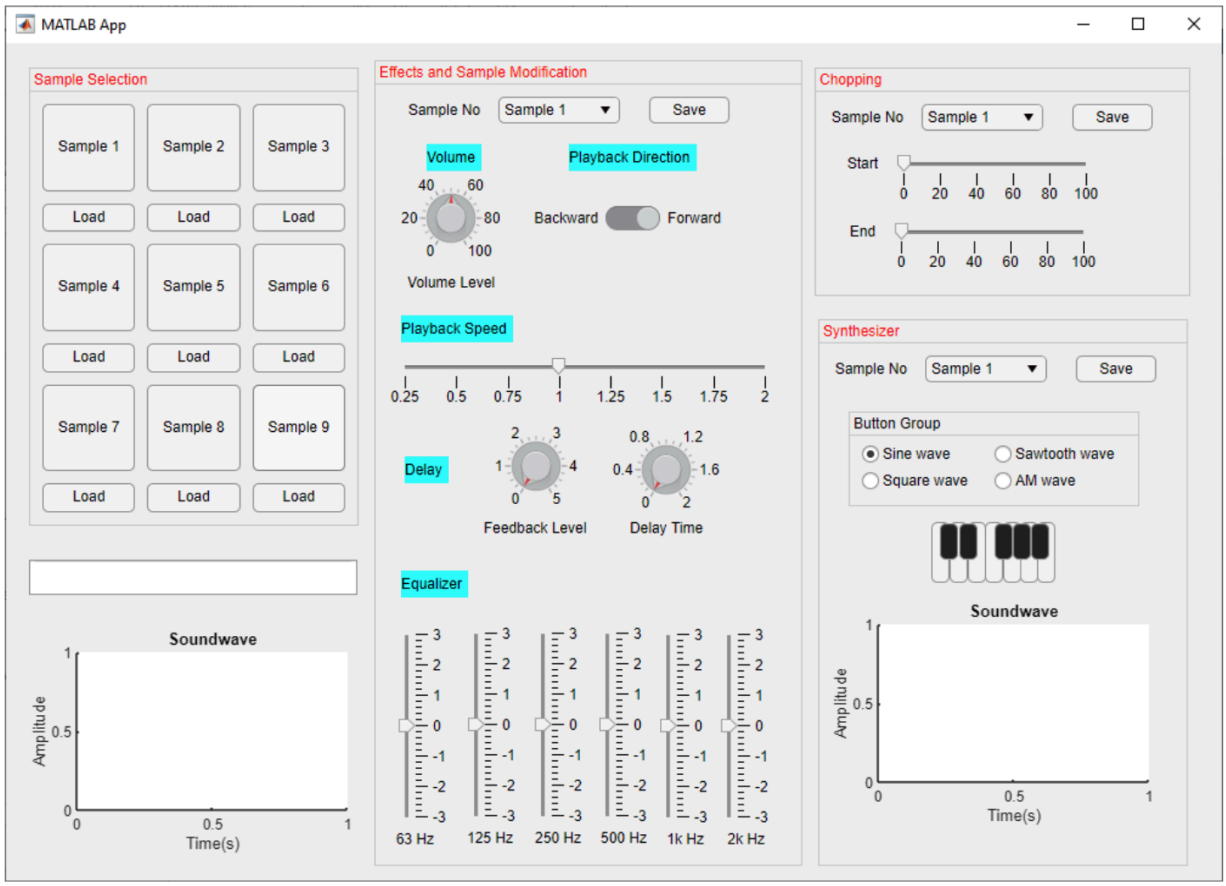

Figure 1: Example of the design view for an audiosampler.